

1 PNSB BSC KPI - Planning Document

PNSB = Permodalan Negeri Selangor Berhad

BSC = Balanced Scorecard (Kad Skor Imbangan)

KPI = Key Performance Indicator (Petunjuk Prestasi Utama)

ALP = Ahli Lembaga Pengarah (Board of Directors)

KPE = Ketua Pegawai Eksekutif (Chief Executive Officer / CEO)

HR = Human Resources (Bahagian Sumber Manusia)

KB = Kakitangan Biasa (Regular/General Staff group — pre-moderation baseline)

KIV = Keep In View (pending — to be confirmed in a later phase)

Full System Name: PNSB BSC KPI (Balanced Scorecard & KPI Performance Management System)

1. Core Concept

A performance management system for PNSB and its subsidiaries that evaluates employees on two pillars — depending on their category:

- **KPI (Quantitative):** *WHAT* was achieved — measured against BSC-aligned targets
- **Penilaian Keperibadian / Competency (Qualitative):** *HOW* they achieved it — personality & behavioural assessment

The **KPI pillar** is structured using the 4 BSC perspectives with fixed company-level weightage:

BSC Perspective	Weightage
💰 Kewangan (Financial)	40%
💛 Pelanggan (Customer)	30%
⚙️ Proses Dalaman (Internal Business Process)	20%
💡 Pembelajaran & Peningkatan (Learning & Growth)	10%

2. Employee Categories & Scoring Split

This is the most critical design decision — scoring formula differs **per employee category**:

Kategori	KPI Weight	Competency (Penilaian Keperibadian) Weight
Ketua Pegawai Eksekutif (CEO)	100%	0%
Eksekutif → Pengurusan Kanan	80%	20%
Sokongan / Teknikal	0%	100%

Key insight: Support/Technical staff are evaluated purely on personality/competency — no KPI scoring.

CEO is purely KPI-driven.

Middle layer (Executive to Senior Management) is the hybrid 80/20 split.

⚠ Important: These scoring splits are formal **ketetapan ALP (ALP Board resolutions)** — not HR or system admin preferences. They can only be changed through a new ALP Board decision. In the system, only Super Admin can modify these values, and only when instructed by an ALP resolution.

3. KPI Target Structure (3-Tier)

Each KPI has **three target levels**, not just one:

Tier	Malay Term	Meaning
Below Threshold	Ambangan (Kurangan)	Minimum acceptable — scored low
Meet Target	Setuju	Hit the agreed target
Stretched	Lebihan	Exceeded target — bonus performance

Example from KPI form (Financial - Revenue KPI, 20% weight):

- Threshold: RM 100,000
- Meet Target: RM 150,000
- Stretched: RM 200,000

KPI scores are computed as: $\text{KPI weight} \times \text{achievement score}$ → summed across all KPIs within each BSC perspective.

4. Process Flow (Proses PMS)

10-step process for PNSB and subsidiaries:

1. **Penyediaan KPI Syarikat** — Prepare Company-level KPI
 2. **Kelulusan KPI Syarikat oleh ALP** — Ahli Lembaga Pengarah (ALP / Board of Directors) approves Company KPI
 3. **Penyediaan KPI Ketua Bahagian / Jabatan / Kakitangan** — Department Heads and staff prepare individual KPIs
 4. **Kelulusan oleh KPE / Ketua Bahagian / Jabatan** — Ketua Pegawai Eksekutif (KPE / CEO) or Department Head approves individual KPIs
 5. **KPI lengkap dikemukakan kepada HR** — Completed KPI forms submitted to Bahagian Sumber Manusia (HR)
 6. **Perlaksanaan KPI (Januari - Disember)** — KPIs are live and tracked for the full year
 7. **KPI lengkap dikemukakan kepada HR untuk penilaian** — Year-end: completed KPIs resubmitted to Bahagian Sumber Manusia (HR) for appraisal
 8. **Sesi penilaian prestasi** — Appraisal session conducted by supervisors/managers, strictly following Ahli Lembaga Pengarah (ALP) formal guidelines and ketetapan (Board resolutions). The scoring splits per employee category (100% KPI / 80%+20% / 100% Competency) are ALP Board decisions — not HR configurations. Any change to these weights requires a new ALP resolution before the system can be updated.
 9. **Sesi Moderasi** — Moderation session by Department Head and Ketua Pegawai Eksekutif (KPE / CEO)
 10. **Kategori Pemarkahan** — Final grade assignment
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5. Performance Rating Categories (Bell Curve)

5-tier grading system (in order, lowest to highest):

#	Kategori	Meaning	Score Range
1	Perlu Diperbaiki	Needs Improvement	< 50
2	Memuaskan	Satisfactory	50 – 59
3	Baik	Good	60 – 75
4	Sangat Baik	Very Good	76 – 89

#	Kategori	Meaning	Score Range
5	Cemerlang	Excellent	90 – 100

Bell Curve Data (actual PNSB data — 91 staff total):

Kategori	Sasaran (Target)	KB	MOD1 (KB/HR)	MOD2 (KPE)	% over target
Perlu Diperbaiki	4	3	0	0	0%
Memuaskan	7	7	17	13	186%
Baik	61	12	54	52	85%
Sangat Baik	16	35	19	24	150%
Cemerlang	3	34	1	2	67%
Jumlah	91	91	91	91	

KB = Kakitangan Biasa (Regular Staff) — raw scores before any moderation

MOD1 (KB/HR) = Moderasi Pertama — after 1st moderation round by Bahagian Sumber Manusia (HR)

MOD2 (KPE) = Moderasi Kedua — after 2nd moderation round by Ketua Pegawai Eksekutif (KPE / CEO)

Bell curve moderation is **confirmed in use** — distribution shifts significantly between rounds

6. Key Modules & Features

User & Organization Management

Organisational Hierarchy (Hierarki Organisasi)

Ahli Lembaga Pengarah - ALP (Board of Directors)

| Sets KPI guidelines & scoring ketetapan. Approves Company KPI.

| Not a system user — decisions are configured by Super Admin.

|

└─ Ketua Pegawai Eksekutif - KPE (Chief Executive Officer)

| Approves all individual KPIs. Leads moderation. 100% KPI scoring.

| Reviewed by: ALP (outside system)

|

└─	Ketua Bahagian / Jabatan (Department Head)
	Approves dept staff KPIs. Participates in moderation.
	Scoring: 80% Petunjuk Prestasi Utama (KPI) + 20% Kompetensi (Competencies)
	Reviewed by: Ketua Pegawai Eksekutif (KPE)
	└─ Eksekutif / Pengurusan Kanan (Executive / Senior Management)
	Submits own KPIs. Self-assesses competencies.
	Scoring: 80% Petunjuk Prestasi Utama (KPI) + 20% Kompetensi (Competencies)
	Reviewed by: Ketua Bahagian
	└─ Kakitangan Sokongan / Teknikal (Support / Technical Staff)
	No KPI form – Kompetensi (Competencies) only.
	Scoring: 100% Kompetensi (Competencies)
	Reviewed by: Ketua Bahagian
└─	Bahagian Sumber Manusia - HR (Human Resources)
	Administers the appraisal cycle. Runs moderation.
	Not in the appraisal chain – system administrator role.

⚠ Exact job titles per category still pending — see **D3** in Questions Tracker.

System Roles & Permissions (Peranan Sistem)

System Role	Maps To	Key Permissions
Super Pentadbir (Super Admin)	System owner / IT	Full access, configure ALP ketetapan (scoring rules), manage all users
Pentadbir Sumber Manusia (HR Admin)	Bahagian Sumber Manusia	Open/close appraisal cycles, run moderation, view all scorecards, lock final grades
Ketua Pegawai Eksekutif (KPE)	CEO	Approve/reject all individual KPIs, lead final moderation round, view company-wide results
Ketua Bahagian (Department Head)	Dept Head	Approve/reject dept staff KPIs, first moderation round, view dept results — also has own scorecard as a reviewee
Kakitangan (Staff)	All other employees	Submit own KPI form, submit self-assessment (Penilaian Kendiri), view own scorecard only

Appraisal Review Chain (Rantai Semakan)

Employee	KPI Approved By	Competency Rated By	Moderation By
Ketua Pegawai Eksekutif (KPE)	Ahli Lembaga Pengarah (ALP) — outside system	— (no competency)	ALP (outside system)
Ketua Bahagian (Dept Head)	Ketua Pegawai Eksekutif (KPE)	KPE rates them	KPE + HR
Eksekutif / Pengurusan Kanan	Ketua Bahagian	Ketua Bahagian rates them	Ketua Bahagian + KPE
Sokongan / Teknikal	— (no KPI form)	Ketua Bahagian rates them	Ketua Bahagian + KPE

Key design note: `manager_id` on the `users` table drives the review chain — the system checks who your manager is to determine who reviews your KPI and competency. A Ketua Bahagian has `manager_id = KPE`. An Eksekutif has `manager_id = Ketua Bahagian`.

KPI Library (BSC-Aligned)

- Company-level KPIs cascaded down to individual level
- KPIs tagged to BSC perspective (Financial / Customer / Internal / Learning)
- Each KPI has 3-tier targets: Threshold / Meet Target / Stretched
- Unit of Measure (UOM) defined per KPI (RM, Date, %, Bil/Count, etc.)

Competency / Penilaian Keperibadian

- Applies to: Executives, Senior Management, Support/Technical staff
- Does NOT apply to: CEO (100% KPI only)
- Sole measure for Support/Technical staff (100% weight)
- Self-rating + Supervisor rating

The Appraisal Workflow

1. Bahagian Sumber Manusia (HR) opens appraisal cycle for the year
2. Company KPI prepared → submitted to Ahli Lembaga Pengarah (ALP / Board of Directors) for approval

3. Staff prepare individual KPIs → submit to Ketua Pegawai Eksekutif (KPE) / Department Head for approval
4. KPIs locked after approval
5. Year runs (Jan–Dec) — staff track and log achievements
6. Year-end: staff submit final KPI achievement data to Bahagian Sumber Manusia (HR)
7. Appraisal session conducted per employee category rules
8. Sesi Moderasi (Moderation session) by Department Head + Ketua Pegawai Eksekutif (KPE) — bell curve applied
9. Final grades assigned and communicated

Dashboards & Analytics (Livewire + hand-rolled SVG charts)

- **Staff Dash:** KPI progress per BSC (Balanced Scorecard) perspective, competency radar chart
- **Ketua Bahagian (Department Head) Dash:** Team KPI completion, distribution preview
- **Bahagian Sumber Manusia (HR) / Admin Dash:** Company-wide bell curve, moderation controls, grade distribution

Audit Trail & Activity Log (*built 2026-06-30*)

- General audit trail via `owen-it/laravel-auditing` — records create / update / delete on all 16 domain & config models (KPIs, BSC perspectives, competencies, departments, designations, cycles, templates, users, scorecards...), capturing old vs new values, the acting user, IP and URL.
- **Authentication auditing** — login, logout, and **failed login** (attempted email + IP) recorded into the same log.
- Secrets (password / remember_token) are never logged; the existing domain logs (`moderation_logs`, `score_overrides`, `scorecard_status_logs`) are left as-is (no double-auditing).
- **Viewer** at `/audit-trail` — **Super Admin only** (separation of duties): filterable table (model / event / actor, with a searchable user combobox) + per-record change-history timeline.

Calculation Preview Tool (Inspired by AIROD PMS+)

HR and Super Admin can plug in scores and preview the computed result before saving — useful during moderation and appraisal sessions.

PNSB Calculation Formula (100% scale — not AIROD's /5 scale):

Step 1 — KPI Score per KPI item:

```
Pencapaian Sebenar = 0 / 60 / 80 / 100 (based on tier achieved)
Skor                  = Pencapaian Sebenar / 100 × Jumlah Kecil Pemberat (KPI item weight)
```

Step 2 — Total KPI Score:

```
KPI Score = SUM(Skor) across all KPI items in the scorecard
```

Step 3 — Competency Score:

```
Competency Score = AVG(manager_rating) across all competency items (method pending B3)
```

Step 4 — Final Score:

```
Final Score = (KPI Score × kpi_weight) + (Competency Score × competency_weight)
```

Example (Eksekutif — 80% KPI / 20% Competency):

```
Final Score = (KPI Score × 0.80) + (Competency Score × 0.20)
```

Step 5 — Grade auto-assigned from Final Score (C5):

```
>= 90 → Cemerlang  
>= 76 → Sangat Baik  
>= 60 → Baik  
>= 50 → Memuaskan  
< 50 → Perlu Diperbaiki
```

The preview tool should show a live simulation panel where HR can adjust individual KPI tier selections and competency scores to see the final score and grade update in real time — before committing the manager override.

7. Database Schema

Full DB design in `db-design.md`. Tables summary:

Table	Purpose
<code>departments</code>	Org structure with parent hierarchy
<code>users</code>	Staff, roles, category, manager_id chain
<code>appraisal_cycles</code>	Annual cycle with status progression
<code>bsc_perspectives</code>	4 fixed BSC perspectives with weights (ALP ketetapan)

Table	Purpose
kpis	KPI library — company / dept / individual level
scorecard_templates	Scoring formula per employee category (ALP ketetapan)
scorecards	One per user per cycle — tracks status, scores, grade
scorecard_kpis	KPI targets & achievements per scorecard
scorecard_competencies	Competency self + manager ratings per scorecard
competency_items	PNSB competency items (Core + Functional)
bell_curve_targets	Target distribution per cycle per grade
moderation_logs	MOD1 / MOD2 grade changes audit
score_overrides	Manager override audit trail
scorecard_status_logs	Full scorecard status change history
audits	General audit trail (owen-it) — create/update/delete on 16 models + login/logout/failed-login; Super Admin viewer

8. Decisions Confirmed

#	Topic	Decision
1	Company Name	PNSB = Permodalan Negeri Selangor Berhad
2	BSC Weights	Financial 40%, Customer 30%, Internal 20%, Learning 10% — fixed
3	Scoring Split	CEO: 100% KPI / Exec→Senior Mgmt: 80% KPI + 20% Competency / Support+Technical: 100% Competency
4	KPI Target Tiers	3-tier: Threshold / Meet Target / Stretched
5	Performance Categories	5 levels: Perlu Diperbaiki / Memuaskan / Baik / Sangat Baik / Cemerlang

#	Topic	Decision
6	Bell Curve Moderation	Confirmed — 2-round moderation (Bahagian Sumber Manusia / HR then Ketua Pegawai Eksekutif / KPE)
7	Appraisal Cycle	Annual — January to December
8	Company KPI Approval	Ahli Lembaga Pengarah (ALP / Board of Directors) approves company KPI
9	Individual KPI Approval	Ketua Pegawai Eksekutif (KPE / CEO) or Department Head approves individual KPIs
10	Role Architecture	Hierarchy-driven (manager_id chain)
11	Rating Scale	Percentage-based (0–100%)
12	Appraisal score method	Manager directly overrides both KPI and Competency scores submitted by employee — no rejection back to employee, no resubmission. Manager edits score in-system. Staff self-rating is reference only. Manager is final authority.

9. ? Questions & Decisions Tracker

Legend: ● **BLOCKER** — must decide before dev starts | ⚠ **NEEDS DECISION** — needed before that module is built | ● **KIV** — deferred to later phase | ✓ **DECIDED** — confirmed and locked

● Blockers (Must Decide First — Pending PNSB HR)

#	Topic	Question	Status
~~B1~~	~~Score → Grade Boundary~~	✓ Resolved — moved to Decided (C5). Grade ranges confirmed from PNSB Excel embedded image.	✓ Done
~~B2~~	~~KPI 3-Tier Score Formula~~	✓ Resolved — moved to Decided (C7). Tier mapping confirmed from PNSB KPI1 Excel data.	✓ Done

#	Topic	Question	Status
B3	Competency Scoring Method	How is self-rating + manager rating combined into one final competency score? See Section 11 for 5 method options with pros & cons.	 Pending
B4	KPI Structure Per Category	Do different employee categories have different KPI sets and BSC structure? Specifically: (1) Do Sokongan/Teknikal staff fill in a KPI form at all, or skip it entirely since their score is 100% Penilaian Keperibadian? (2) Do Eksekutif and KPE share the same KPI items or have different sets? (3) Does the BSC weighting (Financial 40%, Customer 30% etc.) apply to all categories or only those with KPI scoring?	 Pending

Needs Decision (Before That Module Is Built)

#	Topic	Question	Status
D1	Final Sign-off Layers	After moderation, who locks the final grades — HR only, or does KPE also formally sign off? Affects number of approval states in the system.	 TBC
D2	Actual Competency Items	What are PNSB's actual Core and Functional competency items and descriptions? Needed for DB seeding and appraisal form UI.	 TBC
D3	Employee Category Boundaries	Which exact job titles fall under each category (KPE / Eksekutif → Pengurusan Kanan / Sokongan / Teknikal)? Needed to auto-assign the correct scoring template when a user account is created.	 TBC
D4	Subsidiary Handling	Same system shared with anak syarikat, or separate instances? Affects overall multi-tenancy architecture decision upfront.	 TBC

KIV (Deferred — Later Phase)

#	Topic	Note
K1	Bell Curve Hard Enforcement	Hard system cap vs soft guideline — deferred. Phase 1 shows a warning only; hard enforcement scoped to a later phase.
K2	WhatsApp Integration	Submission reminder notifications via WhatsApp — pending scope definition.

✓ Decided (Locked — No Further Action Needed)

#	Topic	Decision
C1	Manager Score Override	Manager directly overrides KPI and Competency scores submitted by employee — no rejection back to employee, no resubmission flow. Manager edits score in-system. Staff self-rating is reference only. Manager is final authority.
C2	Competency Rating Visibility	Manager sees staff self-rating before entering their own score. Both versions saved for audit trail.
C3	ALP Role in Scoring Rules	Scoring splits per category (100% KPI / 80%+20% / 100% Competency) are formal ALP Board ketetapan — not HR or admin configurations. Changes require a new ALP resolution. Only Super Admin can update these in the system.
C4	System Terminology	See Section 10 — terms finalised for all modules and UI labels.
C5	Score → Grade Boundary	Confirmed from PNSB Excel embedded image: Cemerlang 90–100 / Sangat Baik 76–89 / Baik 60–75 / Memuaskan 50–59 / Perlu Diperbaiki < 50. System auto-assigns grade once final score is computed.
C6	Manager Score Override (KPI)	During appraisal, manager directly overrides the KPI score — no dispute or mediation path. Manager is the higher authority; their score is final. Staff cannot contest the manager's rating in-system.
C7	KPI 3-Tier Score Formula	Confirmed from PNSB KPI1 Excel: Threshold (Ambangan) = 60, Meet Target (Setuju) = 80, Stretched (Lebih) = 100, Below Threshold = 0. Hard cap at 100 — exceeding Stretched does not push score above 100.

#	Topic	Decision
		Formula: <code>Skor = (Pencapaian Sebenar / 100) × Jumlah Kecil Pemberat</code> . See <code>files/kpi1-excel-data.md</code> for full verification.

10. Change Impact Notes

Build strategy: Flow and DB design follow current ALP guidelines (locked architecture). These notes flag exactly what changes if a pending decision comes back differently — so we don't re-think from scratch.

If B4 changes — Sokongan/Teknikal also fill KPI form

Current assumption: Sokongan/Teknikal skip KPI form entirely (100% Competency, 0% KPI).

If HR confirms they DO fill KPI form:

What Changes	Impact
<code>scorecard_kpis</code>	Create rows for Sokongan/Teknikal scorecards too — they fill targets & achievements
<code>kpis</code> table	Add <code>category_scope</code> column to assign different KPI sets per category
KPI form routing	Show KPI form to ALL categories, not just KPI-scoring ones
Score calculation	No change — <code>kpi_weight = 0%</code> in their template already handles it
Admin UI	HR/Super Admin needs to manage & assign separate KPI sets per category

DB schema change: Minor — one `category_scope` column on `kpis` table.

Logic change: Medium — form routing + KPI library assignment per category.

If B3 changes — Competency scoring method differs per category

Current assumption: One competency scoring method applies to all categories that have competency (Eksekutif + Sokongan/Teknikal).

If HR confirms different methods per category:

What Changes	Impact
<code>scorecard_templates</code>	Add <code>competency_method</code> column (e.g. <code>weighted</code> , <code>override</code> , <code>average</code>)
Score engine	Branching logic per category when computing <code>final_rating</code> on competency
Config	Method becomes per-template config, not a global setting
UI	No visible change to staff — only affects backend calculation

DB schema change: Minor — one `competency_method` column on `scorecard_templates`.

Logic change: Low — swap formula based on template setting.

If B3 changes — Competency method for Sokongan/Teknikal needs stronger protection

Why: Sokongan/Teknikal are 100% Competency — a biased manager can fully tank their score with no KPI to balance it out.

If HR wants a safeguard for this category specifically:

Option	What to add
Method D (Capped Deviation)	Add deviation check + HR notification logic for Sokongan/Teknikal only
Justification field	Add <code>manager_justification</code> column on <code>scorecard_competencies</code> if deviation exceeds threshold

DB schema change: Minor — one optional `manager_justification` column.

Logic change: Medium — deviation check + HR alert workflow.

11. System Terminology (Finalised)

All UI labels, module names, and field names in the system must follow this convention.

Language rule: Malay label first, English in brackets. Applied consistently across all screens, forms, buttons, and reports.

Core Module Terms

Concept	System Label	Notes
KPI evaluation section	Petunjuk Prestasi Utama (KPI)	Use full Malay name — never just "KPI" alone as a section title
Competency evaluation section	Kompetensi (Competencies)	Use "Competencies" — not "Penilaian Keperibadian" or "Self Assessment" in UI
Balanced Scorecard framework	Kad Skor Imbangan (BSC)	Used as the structural framework label
Overall appraisal form	Borang Penilaian Prestasi (Performance Appraisal Form)	
Appraisal cycle	Kitaran Penilaian (Appraisal Cycle)	
Self rating (by staff)	Penilaian Kendiri (Self Assessment)	Used inside the Competencies section
Manager rating	Penilaian Pengurus (Manager Assessment)	Used inside the Competencies section
Final score	Skor Akhir (Final Score)	
Performance grade	Kategori Prestasi (Performance Grade)	

BSC Perspective Labels

Malay	English (in brackets)
Kewangan	(Financial)
Pelanggan	(Customer)
Proses Dalaman	(Internal Business Process)
Pembelajaran & Peningkatan	(Learning & Growth)

KPI Field Labels

Concept	System Label
KPI target — minimum	Ambangan (Threshold)
KPI target — expected	Sasaran (Meet Target)
KPI target — above expected	Lebihan (Stretched)
Actual achievement	Pencapaian Sebenar (Actual Achievement)
KPI weight (small)	Pemberat Kecil (Sub-weight)
KPI weight (total per BSC)	Pemberat Besar (BSC Weight)
Unit of measure	Unit Pengukuran (Unit of Measure)

Status Labels

Status	System Label
Draft / not submitted	Draf (Draft)
Submitted for review	Dikemukakan (Submitted)
Approved	Diluluskan (Approved)
Rejected	Ditolak (Rejected)
Locked / finalised	Dikunci (Locked)
Under moderation	Dalam Moderasi (Under Moderation)
Completed	Selesai (Completed)

Role Labels (as displayed in UI)

Role	System Label
CEO	Ketua Pegawai Eksekutif (KPE)
Department Head	Ketua Bahagian (Department Head)

Role	System Label
Staff	Kakitangan (Staff)
HR Admin	Pentadbir Sumber Manusia (HR Admin)
Super Admin	Super Pentadbir (Super Admin)

12. Competency Scoring Method

Competency scoring follows **garis panduan dan ketetapan ALP** — the appraisal session is conducted strictly based on ALP Board guidelines and resolutions.

Both ratings (staff self-rating + manager rating) are always saved regardless of outcome — for transparency and audit trail.

*Pending B3 confirmation from PNSB HR. See **Section 10 Change Impact Notes** for what changes in DB and logic if the method differs per category.*

13. Files Reference

File	Description
<code>flow-diagram.md</code>	Full Mermaid flow diagram — all phases + rejection cases
<code>db-design.md</code>	Full database design — all tables, columns, indexes, business rules
<code>files/AIROD_PMS.pdf</code>	Reference PMS slide deck
<code>files/908b7001f2ffdfaf1cc0693c17a88a2.xlsx</code>	PNSB KPI form (KPI1) + Process flow (Proses PMS) — binary
<code>files/kpi1-excel-data.md</code>	Readable export of the Excel above — KPI table, tier scoring formula, B2 resolution proof
<code>files/image.png</code>	Bell curve summary — actual PNSB staff distribution (91 staff)